

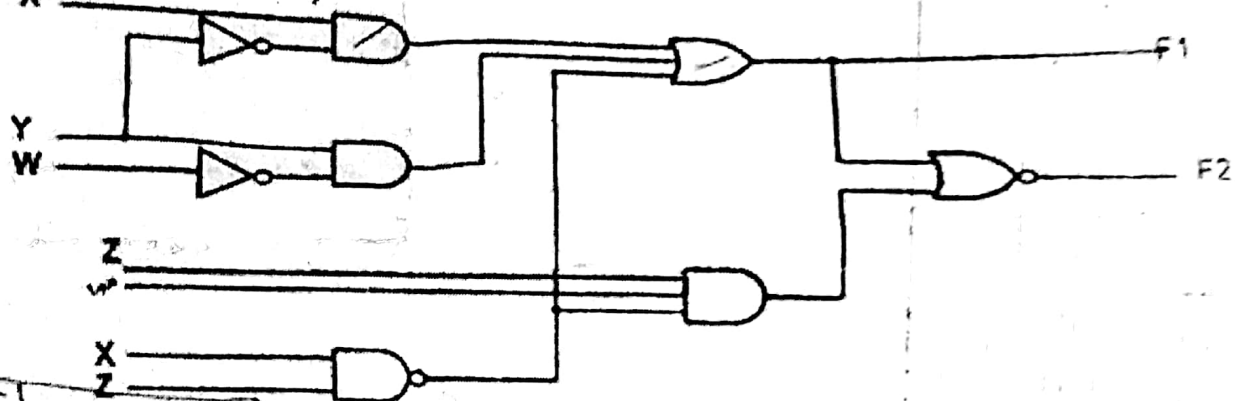
Question No. 1 is Compulsory and
Answer any three questions of the rest

Q1

a. With drawing explain the differences between combinational logic and sequential logic? Give example for each one

b. Implement the following function with multiplexer
 $F(a,b,c,d) = \sum (0,1,2,3,4,8,9,15)$

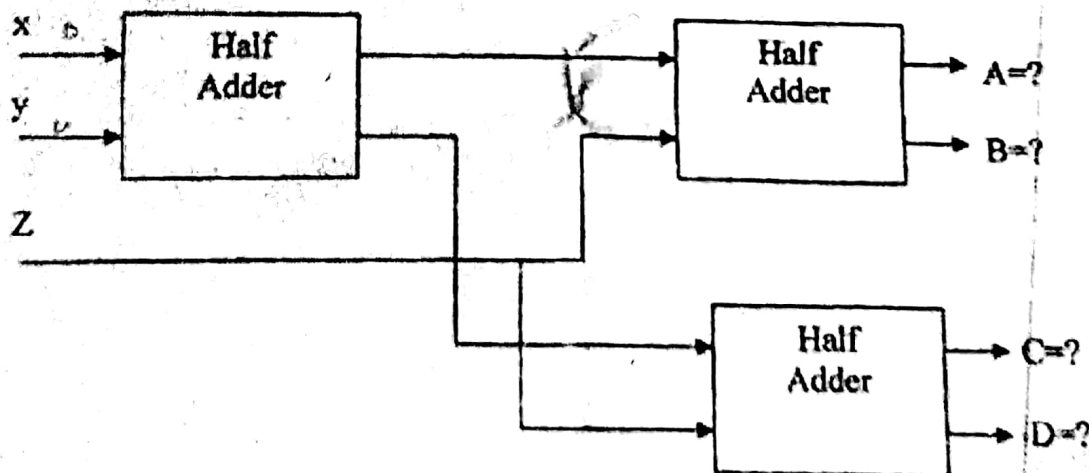
c. Consider the following figure. Find value of F1 and F2? C.



Q2

a. Draw the truth table for a full-adder. Then determine the Boolean Functions for the outputs. Draw a circuit for the full adder using Logical gates.

b. Consider the following figure and answer the following question:

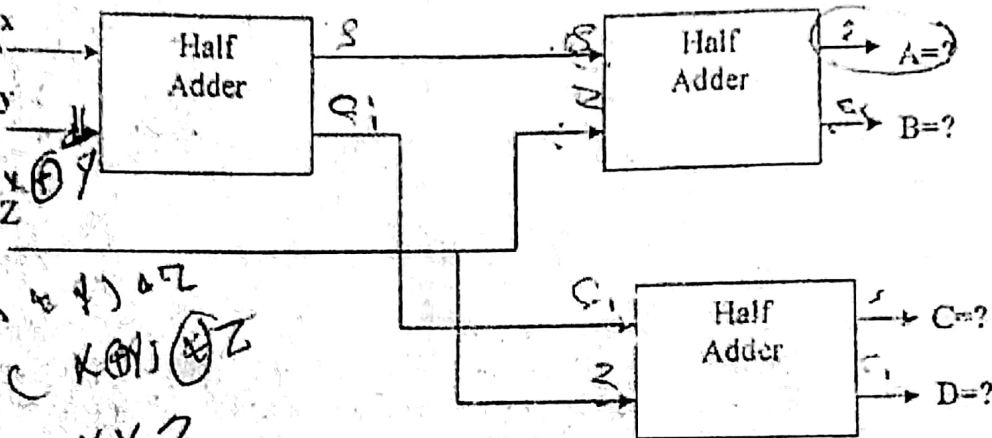


Find out the values of A, B, C, and D?

Note: - Question No.1 is Compulsory.
- Answer any three questions of the rest

Q₁

- a. Draw the truth table for a full-adder: Then determine the Boolean Functions for the outputs. Draw a circuit for the full adder using Logical gates.
- b. Consider the following figure and answer the following question:



Find out the values of A, B, C, and D?

Q₂

Consider the following Boolean function, and answer the following questions:

$$Y = \sum(0, 1, 6, 10, 12, 14)$$

Consider the truth table corresponding to this logical function.
Draw the correct sum-of-product circuit corresponding to this Function.

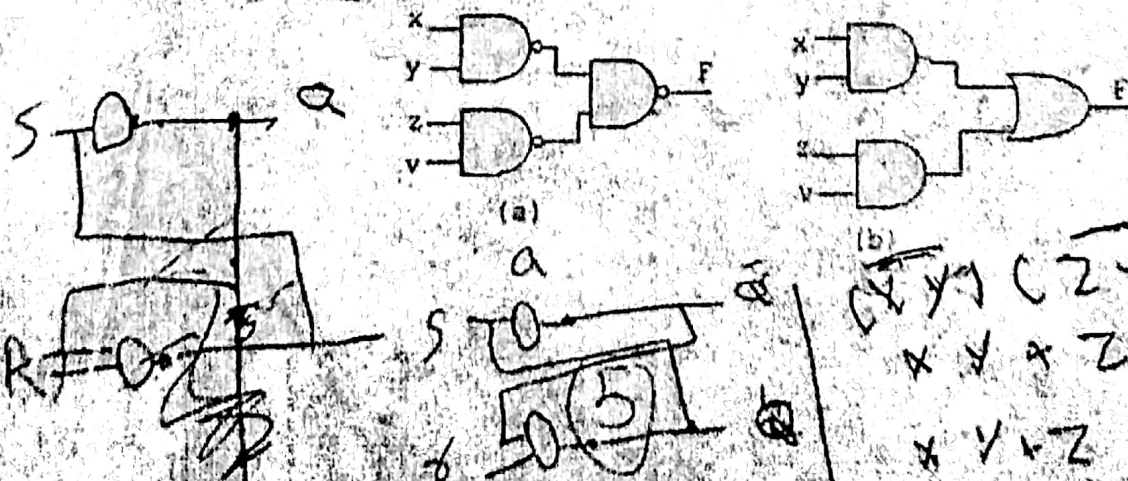
Handwritten truth table and expression:

a	b	c	d	Y
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

$Y = \bar{a}\bar{b}\bar{c} + \bar{a}\bar{b}c + a\bar{b}\bar{c} + a\bar{b}c + a\bar{c} + a\bar{b}c$

Q₃

- a. Draw a circuit diagram for the (unclocked) RS flip-flop and its truth table.
- b. Prove that the circuit fig. (a) And fig. (b) are equivalent.



a. Implement the following function with multiplexer
 $F(a,b,c,d) = \sum M(0,1,3,5,7,10,14)$

b. How do you get Decoder (4×16) by using Decoder (3×8)?

Q₄

Consider the following Truth Table, and answer the following question:

a	b	c	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

- Based on the karnaugh simplification method, draw the sum-of-product circuit corresponding to this function.

Q₅

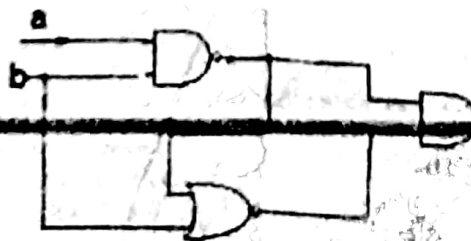
a. Draw the truth table for a full-adder. Then determine the Boolean Functions for the outputs.

b. Construct the truth table for the 3-input EXOR gate

Q₆

Consider the following figure. Then answer the questions:

- Derive the truth table for this circuit
- If $\overline{a} = b$. Find the Boolean function (F)?



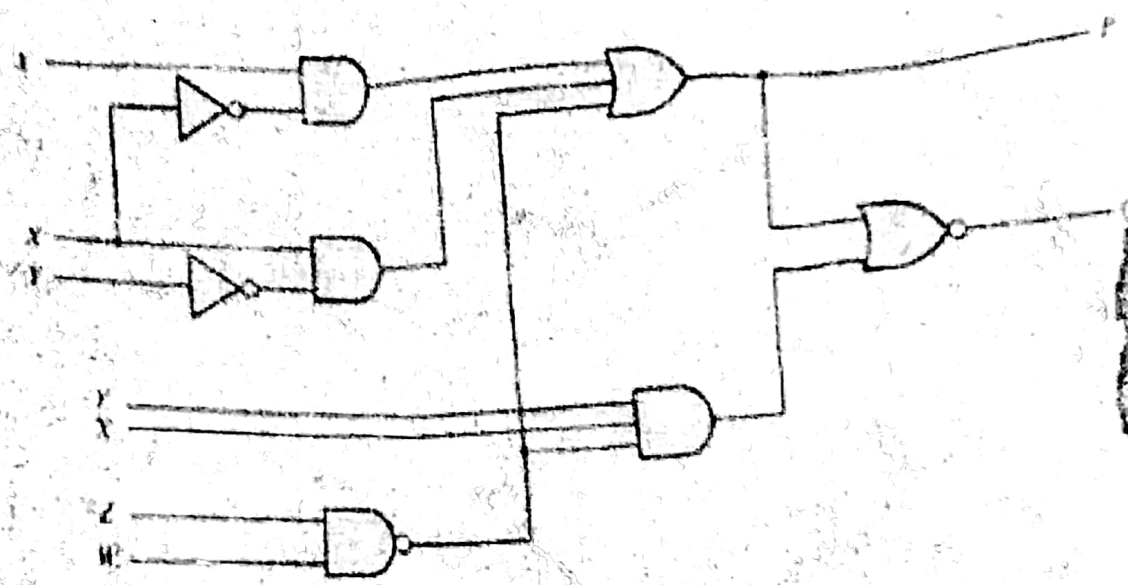
Good luck

Note:

Question No. 1 is Compulsory.

Answer any three questions of the rest

- Q1
- With drawing explain the differences between combinational and sequential logic?
 - Consider the following figure. Find value of P and Q?



- Q2
- What are the differences between asynchronous and synchronous sequential logic? Explain with drawing RS Flip-Flop?

- Combinational logic has 3 inputs (x , y , and z) and 1 output (F). The output equals 1 when the sum of 3-inputs equals 2.
 - Obtain the truth table.
 - Draw the logic diagram.